Diagnostic Analysis of YOLO Architecture

Inductive Bias, Scale Dependency, and Few-Shot Failures in Indian Retail Environments

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Abstract. This report presents a diagnostic analysis of a YOLOv8 object detection model trained on Kirana-Kosh, a custom 22-class dataset of highly dense Indian retail environments. Moving beyond standard deployment metrics, this study treats the model as an object of scientific inquiry to understand its internal mechanisms. Through three controlled experiments, we investigate: (1) how data augmentation acts as an inductive bias to teach low-frequency visual invariances; (2) the architectural dependency of specific product classes on the Feature Pyramid Network (FPN) through targeted ablation; and (3) the exact thresholds where few-shot learning collapses due to class imbalance.

1. Introduction

The unique visual complexity of Indian retail spaces—characterized by densely packed shelves, varied lighting, and high item overlap—presents a significant challenge for computer vision models. Standard evaluations often treat these architectures as black-box tools, prioritizing overall mean Average Precision (mAP) over mechanistic understanding.

This project leverages the Kirana-Kosh dataset (1,189 curated images mapped to 22 Super-Classes) to rigorously probe the behavior of a YOLOv8-Nano architecture. Aligned with the project's mandate for diagnostic inquiry, we frame our research around three sharp questions:

- **Unit I:** Which visual invariances do heavy augmentations actually teach a model when faced with severe image corruption?
- **Unit II:** Which specific Feature Pyramid Network (FPN) detection heads are architecturally critical for parsing specific product scales?
- **Unit III:** At what exact instance threshold does few-shot learning collapse due to intra-class variance in a dense detection setting?

Rather than aiming for state-of-the-art deployment metrics, our objective is to produce interpretation artifacts that answer these questions and explain *why* the model fails under specific constraints.

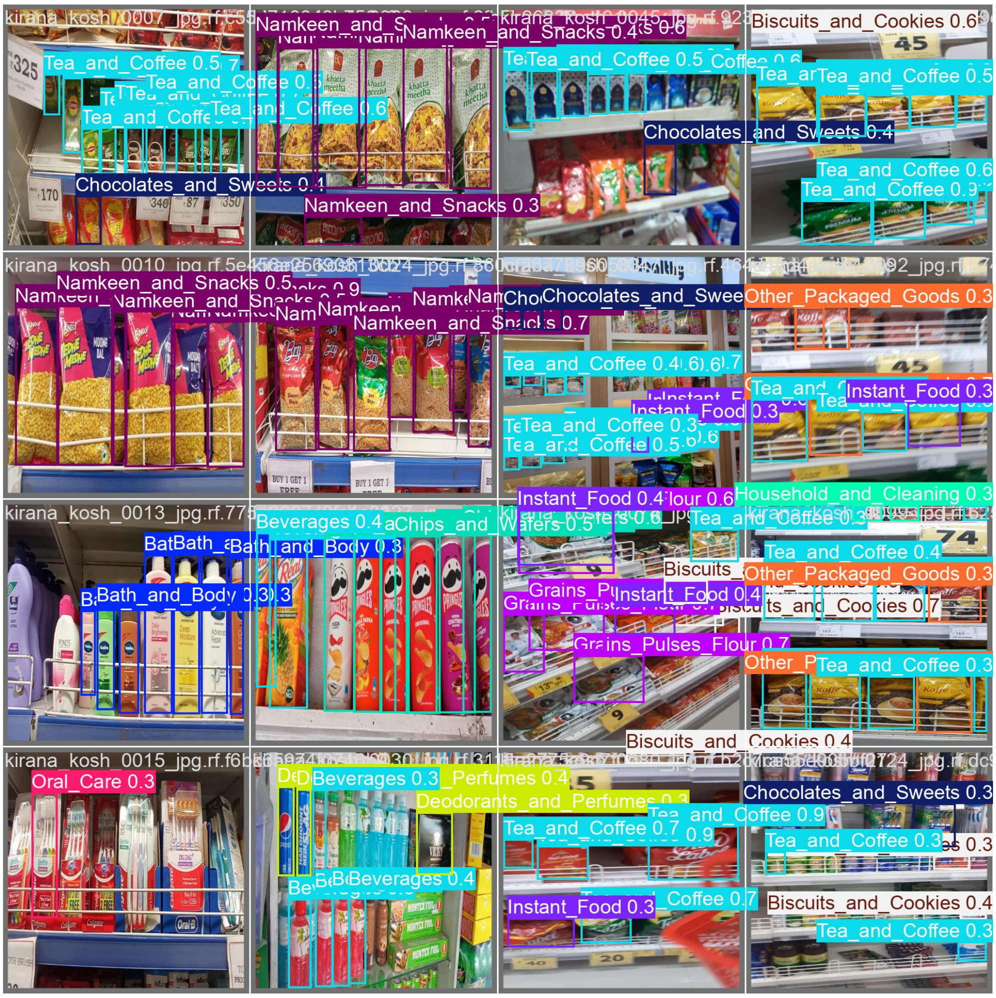

Figure 1: Sample detection from the Kirana-Kosh dataset, highlighting the extreme density, overlapping items, and varied scale that challenge standard object detection paradigms.

2. Unit I: Data Augmentation as Inductive Bias

2.1 Experimental Design

To understand which invariances augmentations teach, we trained two identical YOLOv8-Nano models: a **Baseline** model trained on perfect, untouched images, and an **Augmented** model trained on a dataset multiplied 3x using rotation ($\pm 15°$), brightness variations, and blur. Both models were then evaluated on a deliberately corrupted test set (heavy Gaussian blur applied via OpenCV) to measure robustness.

2.2 Mechanistic Findings

When evaluated on the corrupted data, the Baseline model's mAP50 plummeted to 14.5%, while the Augmented model demonstrated significant robustness, retaining an mAP50 of 24.0%. Specific classes showed even starker contrasts; for instance, *Health_Drinks* dropped to 28.0% on the Baseline but held at 56.8% on the Augmented model.

Why this happens: The Baseline model overfit to "high-frequency" features—sharp text, crisp logo edges, and distinct barcode lines. When the Gaussian blur mathematically destroyed these high-frequency features, the model's predictive capability collapsed, resulting in sparse, low-confidence bounding boxes (as visualized in Figure 3). The Augmented model, however, was repeatedly forced to learn from degraded images. This regularized the model, forcing it to learn *low-frequency, invariant features*—such as the overarching rectangular shape of a box or the relative aspect ratio of a packet. These structural features survive camera blur, proving that augmentation acts as a powerful inductive bias for real-world robustness.

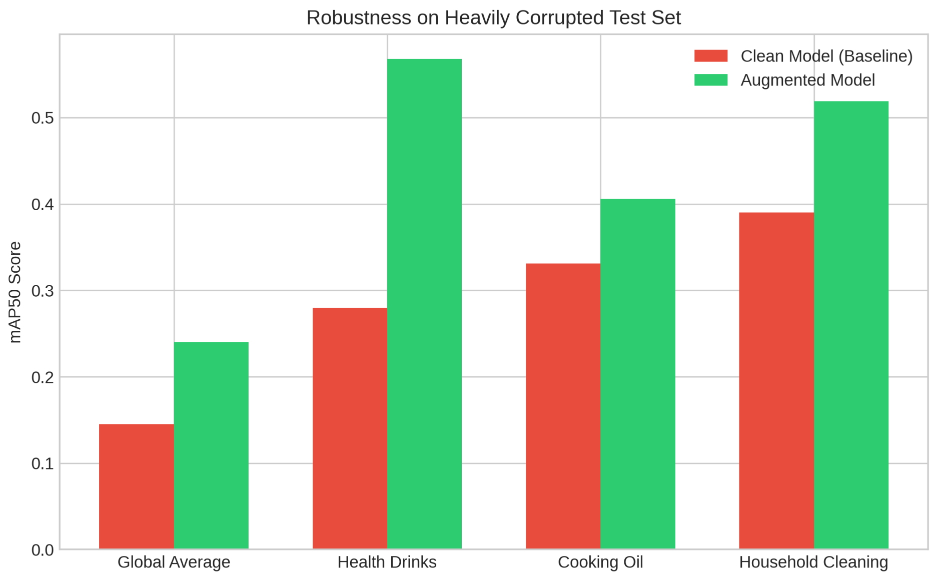

Figure 2: Performance comparison (mAP50) of Baseline vs. Augmented models when evaluated on the heavily corrupted test set.

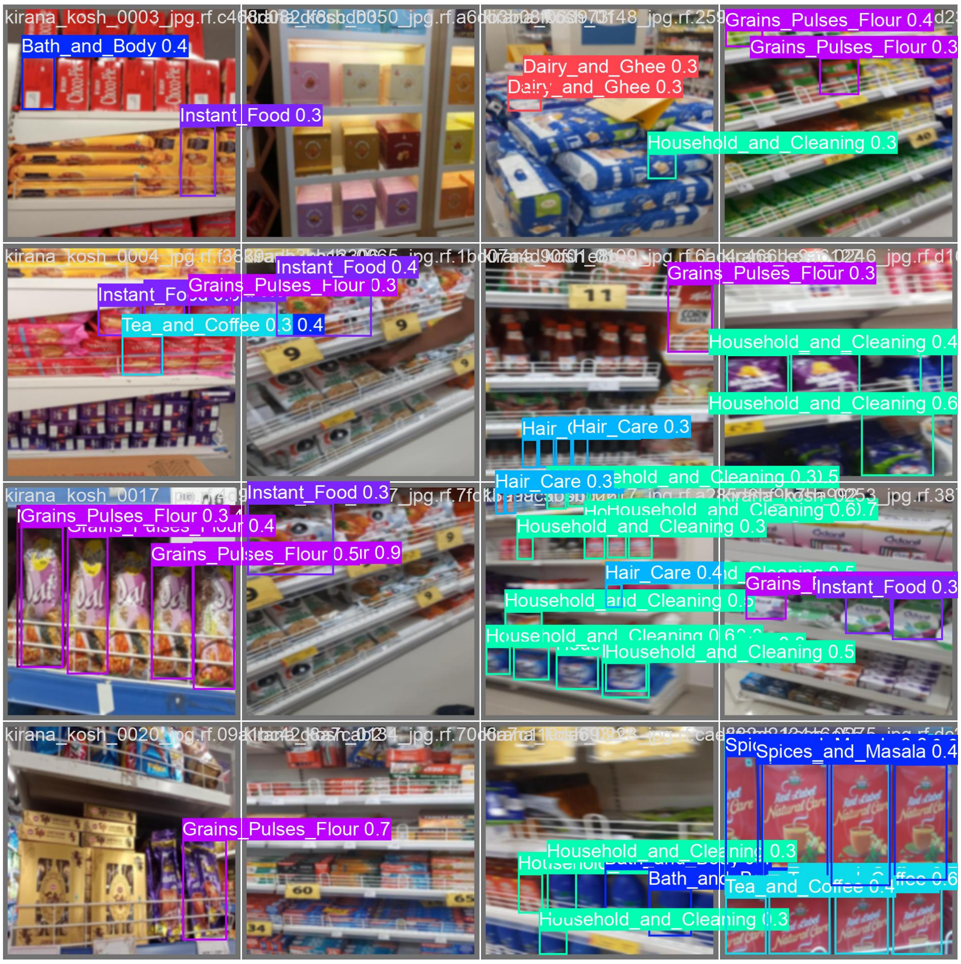

Figure 3: Failure Case (Unit I): The Baseline model failing to detect major items under heavy Gaussian blur, demonstrating its reliance on high-frequency, fragile features rather than structural invariances.

3. Unit II: Feature Pyramid Surgery

3.1 Experimental Design

YOLO architectures utilize a Feature Pyramid Network (FPN) with three distinct detection heads to handle scale variance: P3 (Small objects), P4 (Medium objects), and P5 (Large objects). To understand architectural scale dependency, we performed targeted mathematical ablation on the trained Augmented model. We systematically zeroed out the weights of the `cv2` (bounding box) and `cv3` (classification) branches for one FPN head at a time, running inference to observe class-specific failures.

3.2 Mechanistic Findings

The ablation revealed the network's strict distribution of visual processing:

- **Ablating P3 (Small):** Overall mAP50 dropped from 30.0% to 18.4%. Classes reliant on high-resolution feature maps failed entirely; *Other_Packaged_Goods* (batteries, pens) dropped from 2.3% to 0.6%.
- **Ablating P4 (Medium):** This caused catastrophic failure, with mAP50 plummeting

to 8.2%. Because retail photos are typically taken from a standard "shopping distance," the vast majority of items fall into this pixel area. *Cooking_Oil* collapsed from 52.8% to 1.5%. As clearly seen in Figure 5, the model becomes functionally blind to standard-sized packets on the shelves, entirely skipping rows of prominent inventory.

- **Ablating P5 (Large):** Overall mAP50 dropped to 14.4%. Classes with massive, frame-filling items suffered most, such as *Namkeen_and_Snacks* (family-sized bags), which dropped from 44.9% to 12.5%.

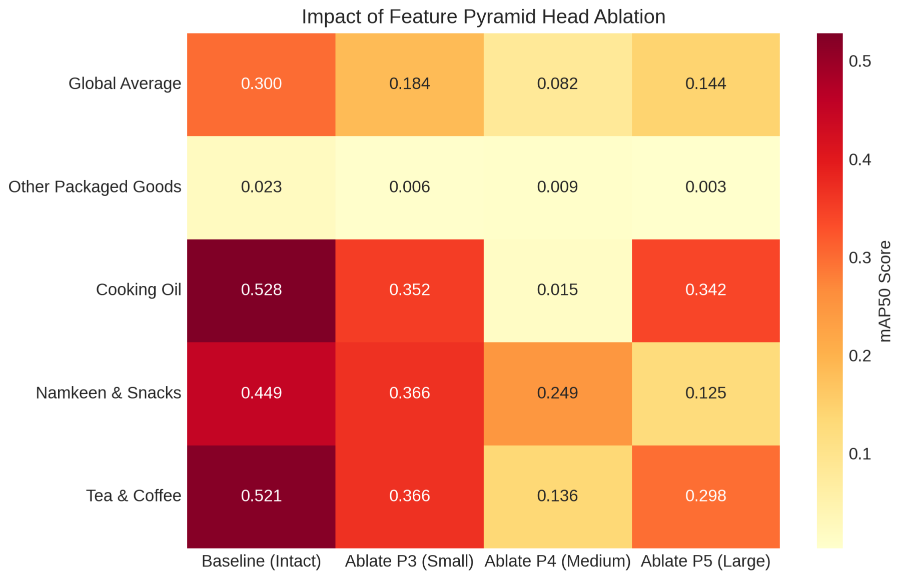

Figure 4: Scale Importance Heatmap: The mAP50 scores across key Super-Classes when specific Feature Pyramid Network (FPN) heads are mathematically ablated.

Figure 5: Failure Case (Unit II): Visualizing detection collapse after P4 (Medium) FPN ablation. The model goes "blind" to standard-sized inventory scales, effectively ignoring entire rows of medium packets while still recognizing items processed by the remaining P3 and P5 heads.

4. Unit III: Few-Shot Learning Failure Modes

4.1 Experimental Design

To map the limits of the architecture's learning capacity, we extracted the per-class Average Precision from the Augmented model's baseline evaluation. By plotting the mAP50 against the exact number of training instances available for that class, we established a taxonomy of few-shot failure in dense detection tasks.

4.2 Mechanistic Findings

The data reveals a clear threshold where few-shot learning breaks down. Classes with extremely low representation, such as *Health_and_Pharma* (30 instances), failed to learn meaningful representations, achieving only ∼3.0% mAP50.

However, the failure taxonomy is not strictly linear; it is heavily influenced by *intra-class variance*. For example, *Chocolates_and_Sweets* possessed 69 instances but still failed (2.3% mAP50). The high visual variance within this Super-Class (ranging from tiny candy bars to large assorted boxes) means that 69 instances are mathematically insufficient to cover the feature distribution space. Conversely, uniform classes required significantly fewer shots to achieve baseline competence.

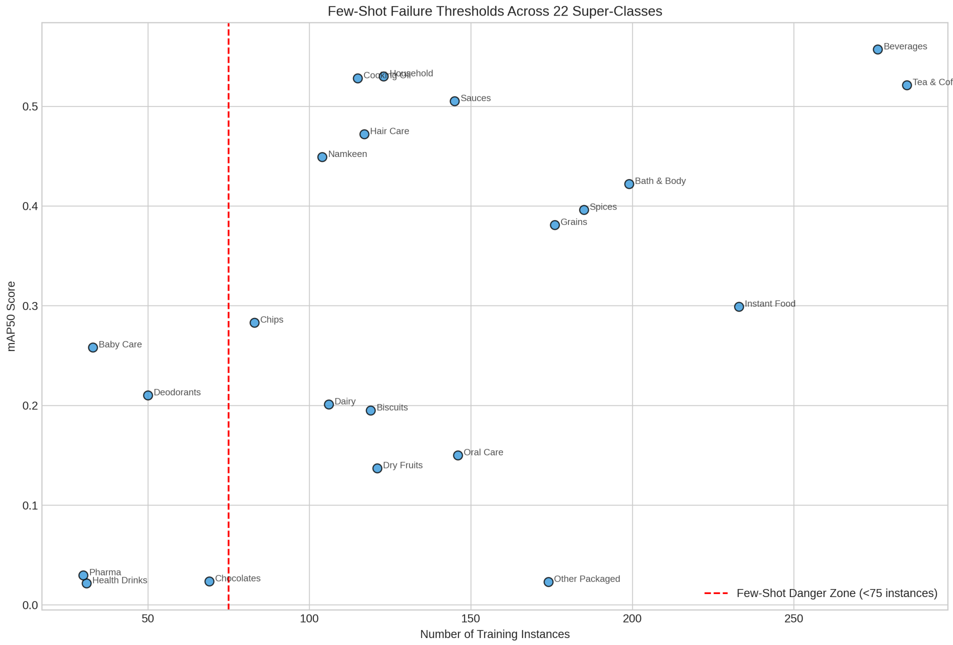

Figure 6: Few-Shot Failure Threshold: Correlation between the number of available training instances and the resulting mAP50 for the 22 Super-Classes.

5. Conclusion

This diagnostic study demonstrates that treating a deep learning model as a scientific subject yields mechanistic insights that standard metric evaluation cannot provide. By probing YOLOv8-Nano trained on the Kirana-Kosh dataset through three controlled experiments, we uncovered the internal logic governing its behavior in dense Indian retail environments.

In Unit I, we proved that data augmentation is not merely a performance booster but a deliberate inductive bias mechanism. Blur and brightness augmentation forced the model to abandon fragile high-frequency cues like sharp text and barcodes, instead learning structurally invariant low-frequency features like packet shape and aspect ratio–features that survive real-world camera degradation.

In Unit II, FPN head ablation revealed that P4 is the architectural backbone of kirana-style retail detection. Because standard shopping-distance photography places the majority of products in the medium pixel range, removing P4 caused catastrophic collapse across nearly all classes–a scale dependency that would be invisible to standard mAP evaluation alone.

In Unit III, we established that few-shot failure in dense detection is not purely a function of instance count but is governed by intra-class visual variance. Chocolates and Sweets failing at 69 instances while uniform classes succeeded at similar counts proves that effective data sufficiency must be measured per class by visual diversity, not raw numbers alone.

Collectively, these findings provide a principled roadmap for future improvement: priority data collection should target high-variance minority classes, FPN architecture choices should be informed by the expected shooting distance of deployment cameras, and augmentation strategies should be treated as domain-specific design decisions rather than default checkboxes.